

FUYUN, China

WMO: 510870

Lat: 46.9941N

Lon: 89.5109E

Elev: 2826

StdP: 13.26

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.2	-22.0	-32.6	1.0	-27.1	-27.3	1.3	-21.9	10.1	8.1	7.6	6.0	0.5	140	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	93.0	61.6	89.9	60.7	87.0	59.9	64.6	81.8	63.2	81.1	62.1	80.3	8.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.0	86.0	68.1	58.1	80.1	67.8	56.1	74.4	68.0	31.2	80.4	30.2	81.2	29.3	80.5	71.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.0	17.5	14.5	DB	-31.2	98.4	7.4	3.6	-36.5	101.0	-40.9	103.2	-45.0	105.2	-50.4	107.8	(4)
(5)			WB	-31.4	66.7	7.4	2.3	-36.7	68.3	-41.0	69.6	-45.2	70.9	-50.5	72.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.7	-0.8	6.8	25.7	47.7	59.4	70.2	73.8	70.3	58.7	44.3	24.4	5.4		
	DBStd	27.82	10.28	10.34	14.20	9.79	8.09	6.17	5.47	6.08	8.18	8.93	13.21	10.35		
	HDD50	6102	1574	1210	758	151	17	0	0	0	20	220	769	1383		
	HDD65	9590	2039	1630	1218	521	209	23	3	21	218	642	1218	1848		
	CDD50	2695	0	0	5	82	308	605	739	630	282	43	1	0		
	CDD65	707	0	0	0	2	35	178	277	185	30	0	0	0		
	CDH74	8036	0	0	0	30	489	1959	2978	2139	432	9	0	0		
	CDH80	3124	0	0	0	4	126	746	1278	860	110	0	0	0		
	WSAvg	4.1	1.2	1.7	3.5	6.1	6.4	6.4	6.0	5.4	4.7	3.7	2.3	1.5		
	PrecAvg	6.6	0.3	0.3	0.4	0.6	0.6	0.6	0.9	0.5	0.6	0.6	0.7	0.5		
(15) Precipitation	PrecMax	10.8	1.0	0.7	1.5	1.2	2.0	2.0	3.0	1.3	2.0	1.4	1.8	1.9		
	PrecMin	3.3	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.1	0.1	0.0	0.0		
	PrecStd	2.1	0.3	0.2	0.4	0.3	0.6	0.5	0.7	0.4	0.5	0.3	0.5	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.4	35.3	64.0	77.8	88.3	94.5	99.5	96.1	88.6	74.9	58.0	33.8		
		MCWB	26.5	28.7	44.5	52.4	57.7	61.7	64.2	62.5	58.5	50.7	41.9	28.2		
	2%	DB	23.2	30.0	57.8	72.7	83.5	90.8	94.0	91.7	82.9	69.2	51.9	28.5		
		MCWB	21.1	25.9	42.4	49.9	56.1	60.6	62.7	60.9	55.8	47.9	39.3	26.1		
	5%	DB	19.0	26.0	51.7	68.4	79.5	87.6	90.2	88.2	78.9	64.5	46.4	24.8		
		MCWB	17.1	22.9	39.4	47.7	54.5	59.6	61.5	60.1	54.3	46.1	36.9	22.5		
	10%	DB	14.8	22.5	46.1	64.5	75.6	84.4	87.0	85.0	75.2	59.7	40.8	20.8		
		MCWB	13.0	20.1	37.9	46.2	52.6	59.0	60.7	59.2	52.9	44.4	34.1	18.8		
	0.4%	WB	26.4	29.3	46.4	54.2	59.2	64.7	67.2	65.6	59.8	52.0	43.5	29.7		
		MCDB	27.8	34.3	59.1	70.8	83.0	81.1	85.0	79.1	81.6	69.1	54.3	32.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	21.4	26.1	43.3	51.8	57.5	63.1	65.4	63.4	57.7	49.4	40.1	26.2		
		MCDB	23.0	29.8	55.3	67.1	78.2	81.3	83.0	82.2	77.3	63.7	50.0	28.5		
	5%	WB	17.2	23.1	40.7	49.8	55.9	61.9	63.9	62.2	55.9	47.4	37.4	22.5		
		MCDB	19.1	25.7	50.2	64.4	75.0	80.6	81.2	81.3	74.5	61.2	45.2	24.6		
	10%	WB	13.0	20.2	37.9	47.7	54.4	60.7	62.8	61.0	54.3	45.6	34.4	18.9		
		MCDB	14.7	22.5	46.1	61.4	71.7	79.2	80.3	79.1	71.4	58.3	40.3	20.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.0	23.4	21.0	21.2	21.5	21.7	21.0	23.1	24.2	21.3	18.3	18.7		
		MCDBR	19.8	23.6	23.1	27.1	27.0	26.8	26.2	28.0	29.1	28.7	25.2	18.4		
		MCWBR	17.6	19.9	12.1	11.4	9.5	8.2	7.5	8.9	11.2	13.8	14.5	15.5		
		MCDBR	19.2	22.2	20.8	23.0	23.4	21.7	20.7	22.3	24.7	25.4	22.8	18.0		
	5% WB	MCWBR	17.1	18.8	11.7	10.5	8.9	6.8	6.5	7.3	9.8	12.8	13.4	15.5		
(40) Clear-Sky Solar Irradiance	taub		0.275	0.309	0.352	0.397	0.386	0.400	0.393	0.371	0.341	0.331	0.301	0.279		
	taud		2.176	2.103	2.113	2.116	2.219	2.254	2.300	2.337	2.373	2.303	2.244	2.156		
	Ebn at Noon		253	267	272	271	279	274	275	277	276	258	241	234		
	Edh at Noon		27	36	42	46	43	42	40	36	32	29	25	24		
(44) All-Sky Solar Radiation	RadAvg		559	881	1321	1768	2138	2256	2187	1933	1544	965	532	411		
	RadStd		40	49	88	92	121	118	102	62	88	77	65	44		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	-1.04	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	-0.72	-1.19	N/A	N/A	N/A	N/A

Nomenclature: See separate page